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F																			
UniSec Medium Voltage Switchgear PT MULTIMAS NABATI ASAHAN MVD-10&MVD-11 20kV Centralized MVD Room SCHEMATIC DIAGRAM DRS RISER WITH VT H10																			
				Date Prep.		15.11.2018		PROJECT TITLE				COVER SHEET		Scale		= H10			
				Prepared		YR		MV Switchgear for MNA Serang Plant				ABB EPMV		Customer		PT Multimas Nabati Asahan & 309			
00		Issued for Approval		28.12.18		YR		Checked		AW		PT. ABB Sakti Industri				Drawing Nbr.		IDUNISEC-BE318042-301 Sh. 1	
Ind.		Revision		Date		Name		Approved		AI		Based on: BE318042				Customer Doc. Nbr.		Cont. 16	
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UniSec

Medium Voltage Switchgear
PT MULTIMAS NABATI ASAHAN
MVD-10&MVD-11 20kV Centralized MVD Room
SCHEMATIC DIAGRAM
DRS RISER WITH VT
H10

ABB EPMV

PT. ABB Sakti Industri

COVER SHEET
MVD-10&MVD-11 20kV Centralized MVD Room
SCHEMATIC DIAGRAM
DRS RISER WITH VT

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A	SHEET INDEX								A					
B	PAGE			PAGE NAME				REVISION						
	==MVD-10_MVD-11=H10&309/1			COVER SHEET				00						
	==MVD-10_MVD-11=H10&309/2			Index of Sheet				00						
	==MVD-10_MVD-11=H10&309/3			REFERENCE DESIGNATIONS				00						
	==MVD-10_MVD-11=H10&309/4			REFERENCE DESIGNATIONS				00						
	==MVD-10_MVD-11=H10&309/5			REFERENCE DESIGNATIONS				00						
	==MVD-10_MVD-11=H10&309/6			REFERENCE DESIGNATIONS				00						
	==MVD-10_MVD-11=H10&309/7			DISTRIBUTION OF AUXILIARY CIRCUITS				00						
	==MVD-10_MVD-11=H10&309/8			MAIN CIRCUITS				00						
	==MVD-10_MVD-11=H10&309/9			LV DOOR LAYOUT				00						
	==MVD-10_MVD-11=H10&309/10			LV BOX LAYOUT				00						
	==MVD-10_MVD-11=H10&309/11			PART LIST				00						
	==MVD-10_MVD-11=H10&309/12			Terminal diagram =H10+-XDI										
	==MVD-10_MVD-11=H10&309/13			Terminal diagram =H10+-XDI2										
	==MVD-10_MVD-11=H10&309/14			Terminal diagram =H10+-XDI6										
	==MVD-10_MVD-11=H10&309/15			Terminal diagram =H10+-XDV										
C	==MVD-10_MVD-11=H10&309/16			Terminal diagram =H10+-XDX							C			
D									D					
E									E					
F					Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	Index of Sheet MVD-10&MVD-11 20kV Centralized MVD Room SCHEMATIC DIAGRAM DRS RISER WITH VT	Scale		= H10
					Prepared	YR	MV Switchgear for MNA Serang Plant					Customer	PT Multimas Nabati Asahan	& 309
00	Issued for Approval	28.12.18	YR	Checked	AW				PT. ABB Sakti Industri	Drawing Nbr.		IDUNISEC-BE318042-301	Sh.	2
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042			Customer Doc. Nbr.		Cont.	16	
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A	REFERENCE DESIGNATION OF OBJECTS IN ELECTRICAL DOCUMENTS				5		6		7		8		A
	(IN COMPLIANCE WITH STANDARD IEC 81346-2 AND ABB TECHNICAL STANDARD 2NBA000001)												
B	DESIGNATION	DESCRIPTION											B
	-AA	MULTIFUNCTION UNIT (CENTRAL UNIT)											
	-BAD1,-BAD4	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L1											
	-BAD2,-BAD5	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L2											
	-BAD3,-BAD6	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L3											
	-BAR	VOLTAGE PROTECTION RELAY											
	-BAS1	VOLTAGE SENSOR LOCATED ON PHASE L1											
	-BAS2	VOLTAGE SENSOR LOCATED ON PHASE L2											
	-BAS3	VOLTAGE SENSOR LOCATED ON PHASE L3											
	-BAT1	VOLTAGE TRANSFORMER LOCATED ON PHASE L1											
C	-BAT2	VOLTAGE TRANSFORMER LOCATED ON PHASE L2											C
	-BAT3	VOLTAGE TRANSFORMER LOCATED ON PHASE L3											
	-BAT4	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L1											
	-BAT5	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L2											
	-BAT6	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L3											
	-BCD	DIFFERENTIAL PROTECTIVE RELAY											
	-BCF	FEEDER PROTECTION RELAY											
	-BCG	GENERATOR PROTECTION RELAY											
	-BCM	MOTOR PROTECTION RELAY											
	-BCN	NEUTRAL (RESIDUAL) CURRENT TRANSFORMER											
D	-BCP	TRANSFORMER PROTECTION RELAY											D
	-BCR	CURRENT PROTECTION RELAY											
	-BCS1	CURRENT SENSOR LOCATED ON PHASE L1											
	-BCS2	CURRENT SENSOR LOCATED ON PHASE L2											
	-BCS3	CURRENT SENSOR LOCATED ON PHASE L3											
	-BCT1	CURRENT TRANSFORMER LOCATED ON PHASE L1											
	-BCT2	CURRENT TRANSFORMER LOCATED ON PHASE L2											
	-BCT3	CURRENT TRANSFORMER LOCATED ON PHASE L3											
	-BCT4,-BCT7	CURRENT TRANSFORMER LOCATED ON PHASE L1											
	-BCT5,-BCT8	CURRENT TRANSFORMER LOCATED ON PHASE L2											
E	-BCT6,-BCT9	CURRENT TRANSFORMER LOCATED ON PHASE L3											E
	-BCZ	DISTANCE PROTECTION RELAY											
	-BEF	FREQUENCY PROTECTION RELAY											
	-BER	SUPERVISION RELAYS											
	-BES	SYNCHRONIZING RELAY											
	-BET	THERMAL PROTECTION RELAY											
	-BGA1	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CIRCUIT-BREAKER COMPARTMENT											
	-BGA2	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN BUSBAR COMPARTMENT (SINGLE BUSBAR SYSTEM) OR IN FRONT BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)											
	-BGA3	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CABLE COMPARTMENT											
	-BGA4	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN REAR BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)											
F	-BGB1...-BGB3	POSITION SWITCHES OF CIRCUIT-BREAKER											F
	-BGB4	POSITION SWITCH OF CIRCUIT-BREAKER: PASSING CONTACT CLOSING MOMENTARILY WHEN THE BREAKER OPENS											

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A	-BGI31	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION						-CA	CAPACITORS						A	
	-BGI32	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION						-CA1	CAPACITOR FOR FRONT VENTILATOR							
	-BGI33	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)						-CA2	CAPACITOR FOR REAR VENTILATOR							
	-BGI41...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION						-EA1	LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.							
	-BGI42	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION						-EA2	LIGHTING LAMP LOCATED IN CABLE COMPARTMENT							
	-BGI43...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION						-EA3	LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE							
	-BGI51	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION						-EA4	LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT							
	-BGI52	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION						-EB1	HEATER LOCATED IN CABLE COMPARTMENT							
	-BGI53	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION						-EB2	HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT							
	-BGK	POSITION SWITCH OF KEY LOCK						-EB3	HEATER LOCATED IN MOTOR CABINET							
B	-BGL1	POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1						-EB4	HEATER FOR VENTILATION TEST						B	
	-BGS1...-BGS4	POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS						-EB5	HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.							
	-BGS5	PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION						-EB7	HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE							
	-BGT1	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION						-EB8	HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE							
	-BGT2	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION						-FA1	SURGE ARRESTER LOCATED ON PHASE L1							
	-BGT3	POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION						-FA2	SURGE ARRESTER LOCATED ON PHASE L2							
	-BGT4	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION						-FA3	SURGE ARRESTER LOCATED ON PHASE L3							
	-BGT5	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION						-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS							
	-BGT6	POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION						-FCF1	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1							
	-BGT7	POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION						-FCF2	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2							
C	-BGT8	POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION						-FCF3	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3						C	
	-BGT11	PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION						-FCM1	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER							
	-BGT12	PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION						-FCM2....-FCM10	MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS							
	-BGV1	POSITION SWITCH OF FRONT VENTILATOR						-FCM11	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1							
	-BGV2	POSITION SWITCH OF REAR VENTILATOR						-FCM12	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2							
	-BM	HYGROSTAT						-FCM13	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3							
	-BPA4	PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-FCT	THERMAL OVERLOAD RELEASES							
	-BPA5	PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GA	GENERATORS							
	-BPA6	PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ1	FRONT VENTILATOR							
	-BPA7	PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ2	REAR VENTILATOR							
D	-BPS1	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER						-GQ3	VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK						D	
	-BPS2	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2						-KFA1	AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE							
	-BPS3	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3						-KFA2	AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE							
	-BR	FLAME DETECTORS, SMOKE DETECTORS						-KFA3...-KFA9	AUXILIARY RELAYS OR CONTACTORS							
	-BT	THERMOSTAT						-KFC	CLOSING RELAYS OR CONTACTORS							
	-BT1	THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS						-KFI	INTEGRATED CIRCUITS							
	-BT2	THERMOSTAT FOR ENERGIZATION OF VENTILATORS						-KFL	LOCKOUT RELAY							
	-BT3	THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM						-KFM	MICROPROCESSORS							
	-BU	BUCHHOLZ RELAY						-KFN	ANTIPUMPING RELAYS							
	-BX1	UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING						-KFO	OPENING RELAYS OR CONTACTORS							
E	-BX2	ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING						-KFP	PROGRAMMABLE LOGIC CONTROLLERS (PLC)						E	
	-BZ,-BZ4	MULTIFUNCTION SENSORS						-KFR	RECLOSING RELAYS							
	-BZ1	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1						-KFS	SYNCHRONIZING DEVICES							
	-BZ2	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2						-KFT	AUXILIARY TIME RELAYS, DELAY ELEMENTS							
	-BZ3	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3						-KFZ	CONTROLLERS							
								-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER							
								-MAD	MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD							
								-MAD1	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1							
								-MAD2	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2							
								-MAD9	MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS							
F					Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB	EPMV	REFERENCE DESIGNATIONS MVD-10&MVD-11 20kV Centralized MVD Room		Scale		= H10	F
					Prepared	YR							Customer	PT Multimas Nabati Asahan	& 309	
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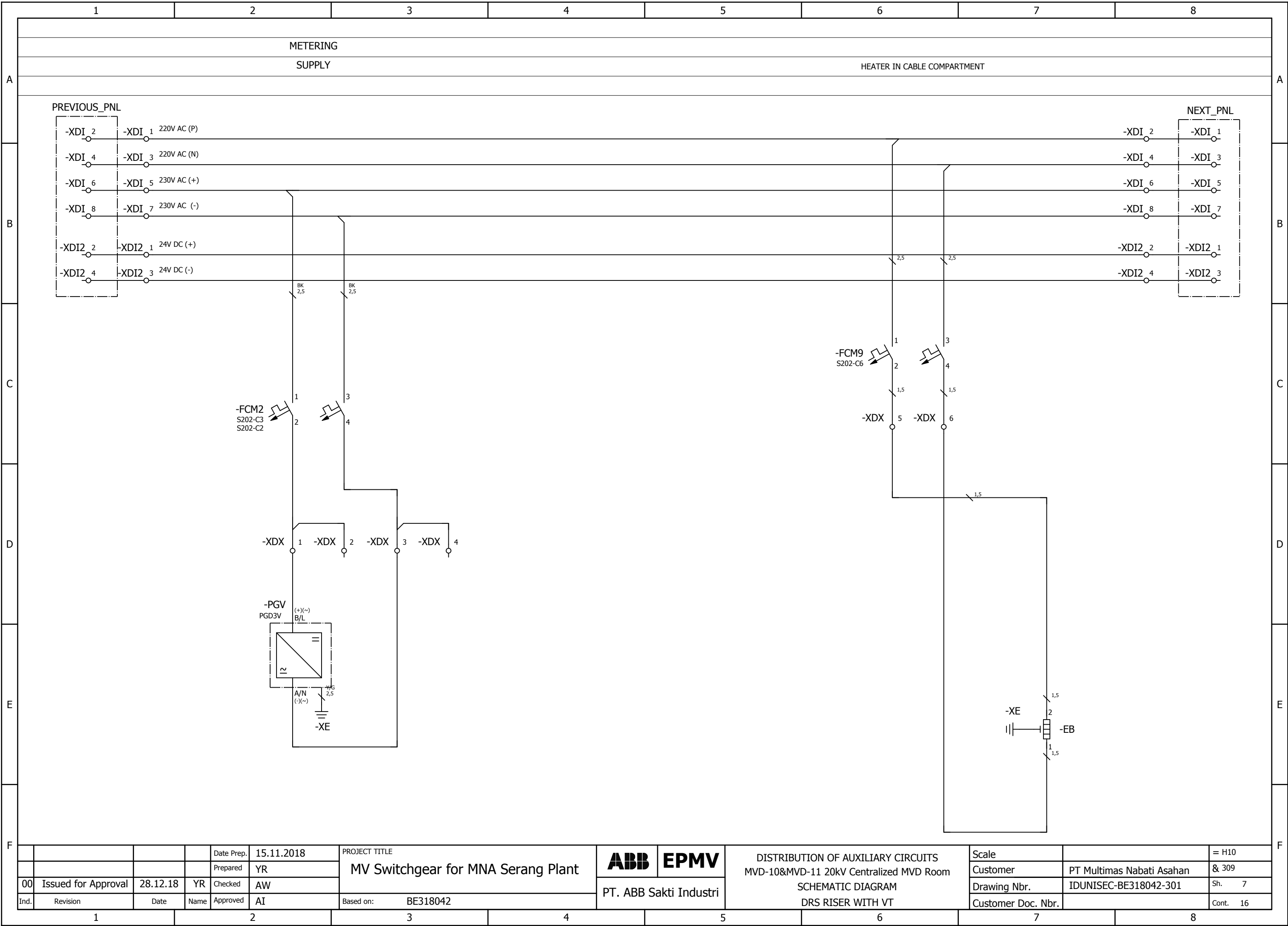
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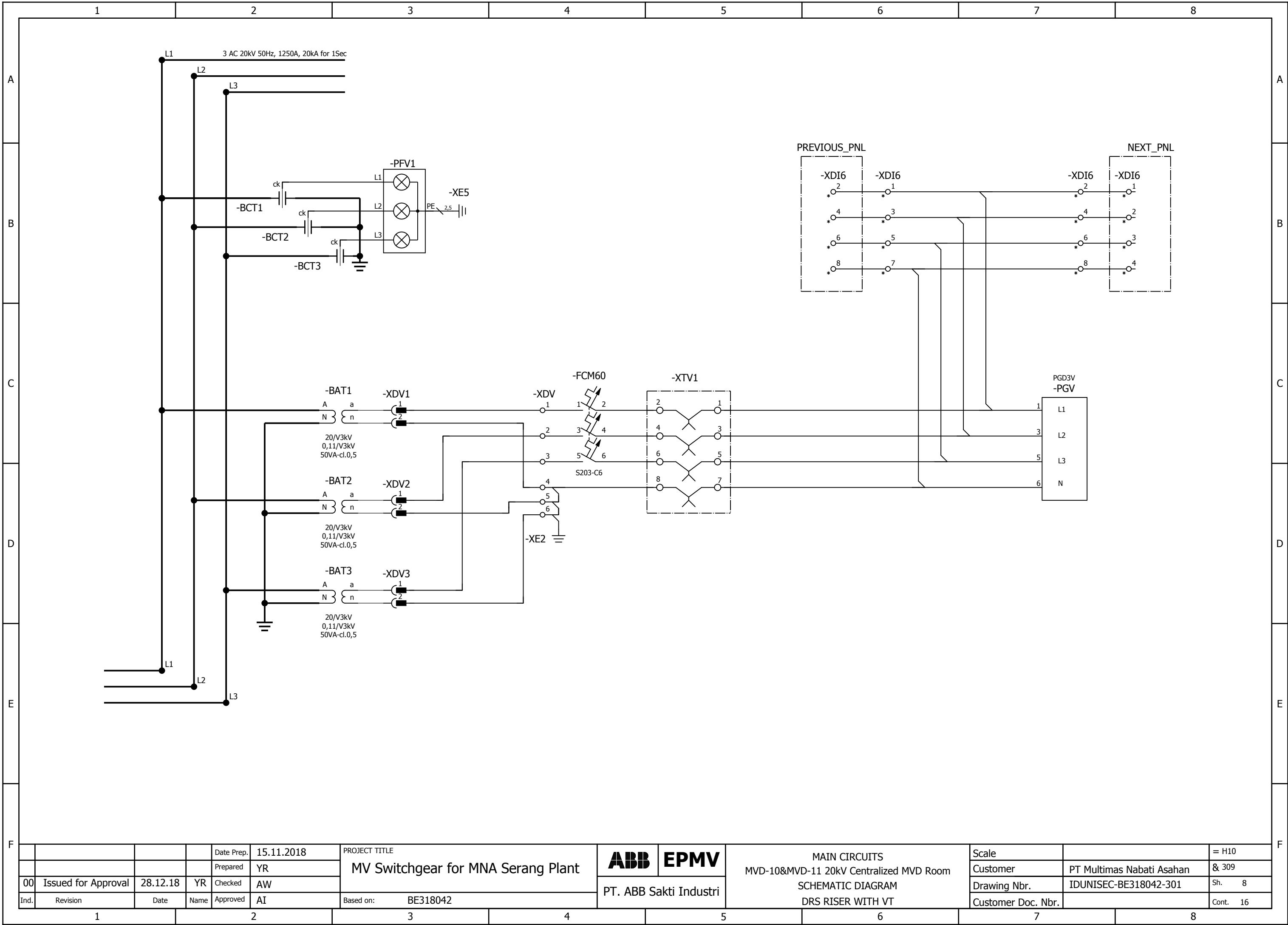
A	1	2			3			4			5		6		7		8																	
	-MAE		MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE															-QBH		MANUAL CIRCUIT-BREAKERS														
	-MAE1		MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2															-QBL		LINKS														
	-MAE8		MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS															-QBM		MINIATURE SWITCH-DISCONNECTORS														
																		-QBS		SWITCH-DISCONNECTORS														
																		-QBT		TRUCK														
																		-QCA		EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING														
																		-QCE		EARTHING SWITCH														
																		-QCE1		EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)														
B	-MBO1		FIRST OPENING RELEASE OF CIRCUIT-BREAKER															-QCE2		EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)														
	-MBO2		SECOND OPENING RELEASE OF CIRCUIT-BREAKER															-QQ		CLUTCHES														
	-MBO3		OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER															-RAA		ANTIFERRORESONANCE RESISTOR														
	-MBO4		OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS															-RAD		DIODES														
	-MBU		UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER															-RAI		INDUCTORS														
	-PFB		BLUE SIGNAL LAMPS															-RAR		RESISTORS														
	-PFF		FLAG RELAYS															-RF		FILTERS														
	-PFG		GREEN SIGNAL LAMPS															-RLE1		ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING														
	-PFR		RED SIGNAL LAMPS															-RLE2		ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT														
	-PFS		SHORT CIRCUIT INDICATORS															-RLE3		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE														
C	-PFV		VOLTAGE INDICATORS																															
	-PFV1		VOLTAGE INDICATORS ON FEEDER SIDE															-RLE4		ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION														
	-PFV2		VOLTAGE INDICATORS ON BUSBAR SIDE															-RLE5		ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD														
	-PFW		WHITE SIGNAL LAMPS															-RLE6		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1														
	-PFX		CROSS INDICATORS, ELECTROMECHANICAL INDICATORS																															
	-PFY		YELLOW SIGNAL LAMPS															-RLE7		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2														
	-PGA		AMMETERS															-RLE8		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2														
	-PGC		COUNTERS																															
	-PGF		FREQUENCYMETERS															-RLE9		ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS														
	-PGH		HOURMETERS															-RLE11		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4														
D	-PGI		PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE															-RLE12		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5														
	-PGI1		HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)																															
	-PGI2		HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)															-SFA		AMMETRIC SWITCHES														
	-PGJ		ACTIVE ENERGY METERS															-SFC		CONTROL SWITCHES, CLOSING PUSH-BUTTONS														
	-PGK		REACTIVE ENERGY METERS															-SFL		LOCKING CONTACTS														
	-PGM		MULTIFUNCTION INDICATORS															-SFO		OPENING PUSH-BUTTONS														
	-PGP		POWER-FACTOR METERS															-SFR		RESET PUSH-BUTTONS														
	-PGQ		VARMETERS															-SFS		SELECTOR SWITCHES														
	-PGS		SYNCHRONOSCOPES															-SFT		TEST PUSH-BUTTONS														
	-PGV		VOLTMETERS															-SFU		UNLOCKING PUSH-BUTTONS														
E	-PGW		WATTMETERS															-SFV		VOLTMETRIC SWITCHES														
	-PJ		ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)															-TA		POWER TRANSFORMERS														
	-QAB		CIRCUIT-BREAKERS															-TB		CONVERTER														
	-QAC		CONTACTORS (FOR POWER)															-TB1		RECTIFIER FOR FIRST OPENING RELEASE														
	-QBD		DISCONNECTORS															-TB2		RECTIFIER FOR SECOND OPENING RELEASE														
	-QBD1		FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)															-TB3		RECTIFIER FOR CLOSING RELEASE														
	-QBD2		REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)															-TB4		RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING														
	-QBD3		FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)															-TB5		RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT														
	-QBD4		TWO-WAY DISCONNECTOR															-TB6		RECTIFIER FOR UNDERVOLTAGE RELEASE														
	-QBD5		TWO-WAY DISCONNECTOR ON BUSBAR RISER															-TB7		ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER														
F					Date Prep.	15.11.2018	PROJECT TITLE				ABB	EPMV	REFERENCE DESIGNATIONS				Scale		= H10															
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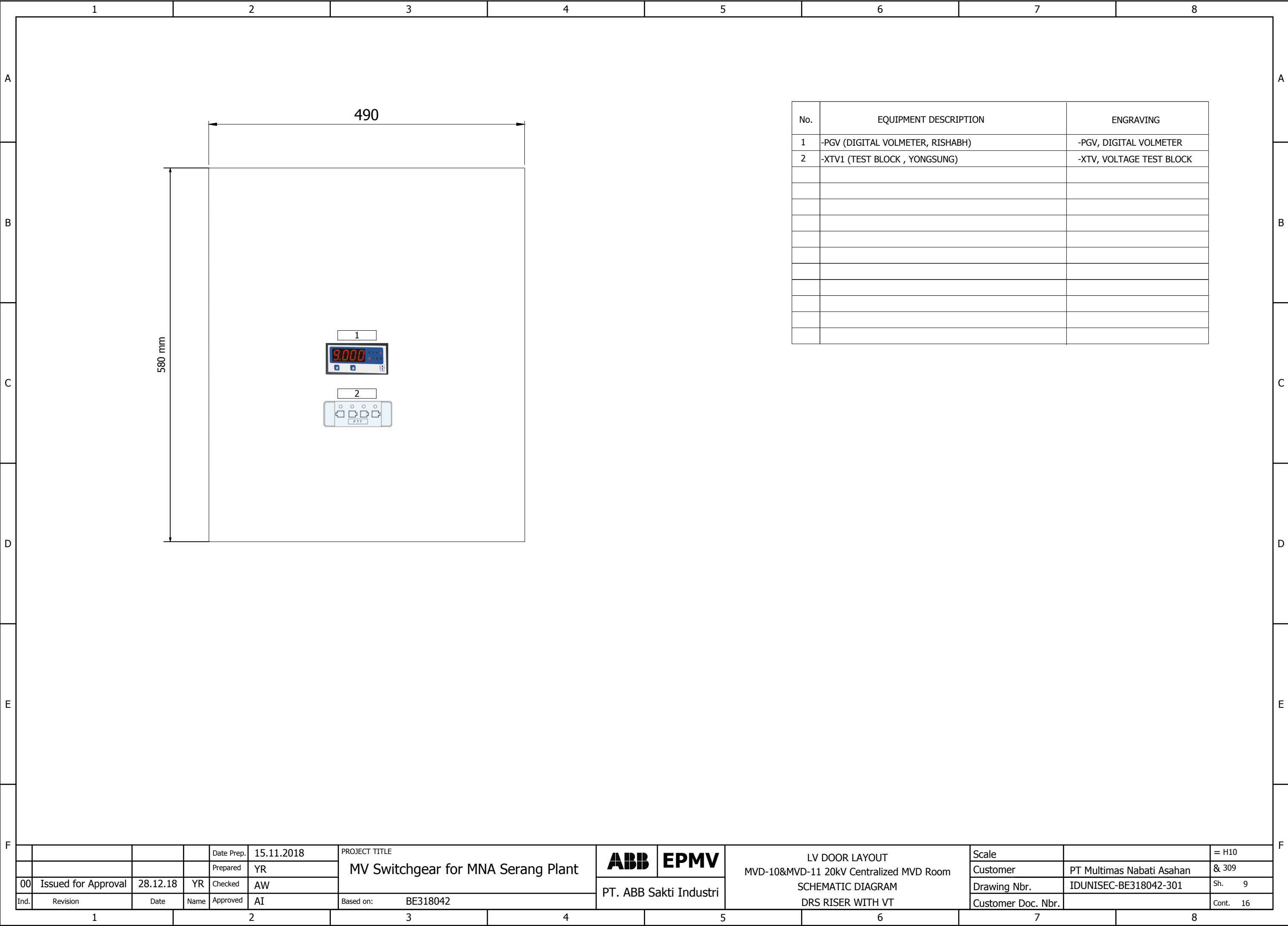
A	-TFA	ACTIVE POWER TRANSDUCERS	A
	-TFC	CURRENT TRANSDUCERS	
	-TFF	FREQUENCY TRANSDUCERS	
	-TFJ	ACTIVE ENERGY TRANSDUCERS	
	-TFK	REACTIVE ENERGY TRANSDUCERS	
	-TFM	MULTIFUNCTION TRANSDUCERS	
	-TFP	POWER-FACTOR TRANSDUCERS	
	-TFQ	REACTIVE POWER TRANSDUCERS	
	-TFS	SIGNAL CONVERTERS	
	-TFV	VOLTAGE TRANSDUCERS	
B	-WA	BUSBARS	B
	-WBC	POWER CABLES	
	-WE	EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.	
	-WGC	CONTROL CABLES	
	-WGS	COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)	
	-WH	OPTICAL FIBRES	
	-XDA	TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS	
	-XDA1	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1	
	-XDA2	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2	
	-XDA3	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3	
C	-XDA5	CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER	C
	-XDB	CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS	
	-XDB1	CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)	
	-XDB2	CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER	
	-XDB10...-XDB89	CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS	
	-XDB7	CONNECTOR FOR TRUCK POSITION CONTACTS	
	-XDB91	CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR	
	-XDB92	CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR	
	-XDB93	CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR	
	-XDB95	CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET	
D	-XDC	CUSTOMER TERMINAL BLOCK	D
	-XDC1	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER	
	-XDC2	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR	
	-XDC3	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT	
	-XDC4	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS	
	-XDC5	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER	
	-XDE	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE	
	-XDE1	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE	
	-XDE2	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2	
	-XDE3	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2	
E	-XDF	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS	E
	-XDF1	CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR	
	-XDF2	CONNECTOR FOR REAR VENTILATOR CIRCUITS	
	-XDF3	CONNECTOR FOR FRONT VENTILATOR CIRCUITS	
	-XDH	TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)	
	-XDI	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)	

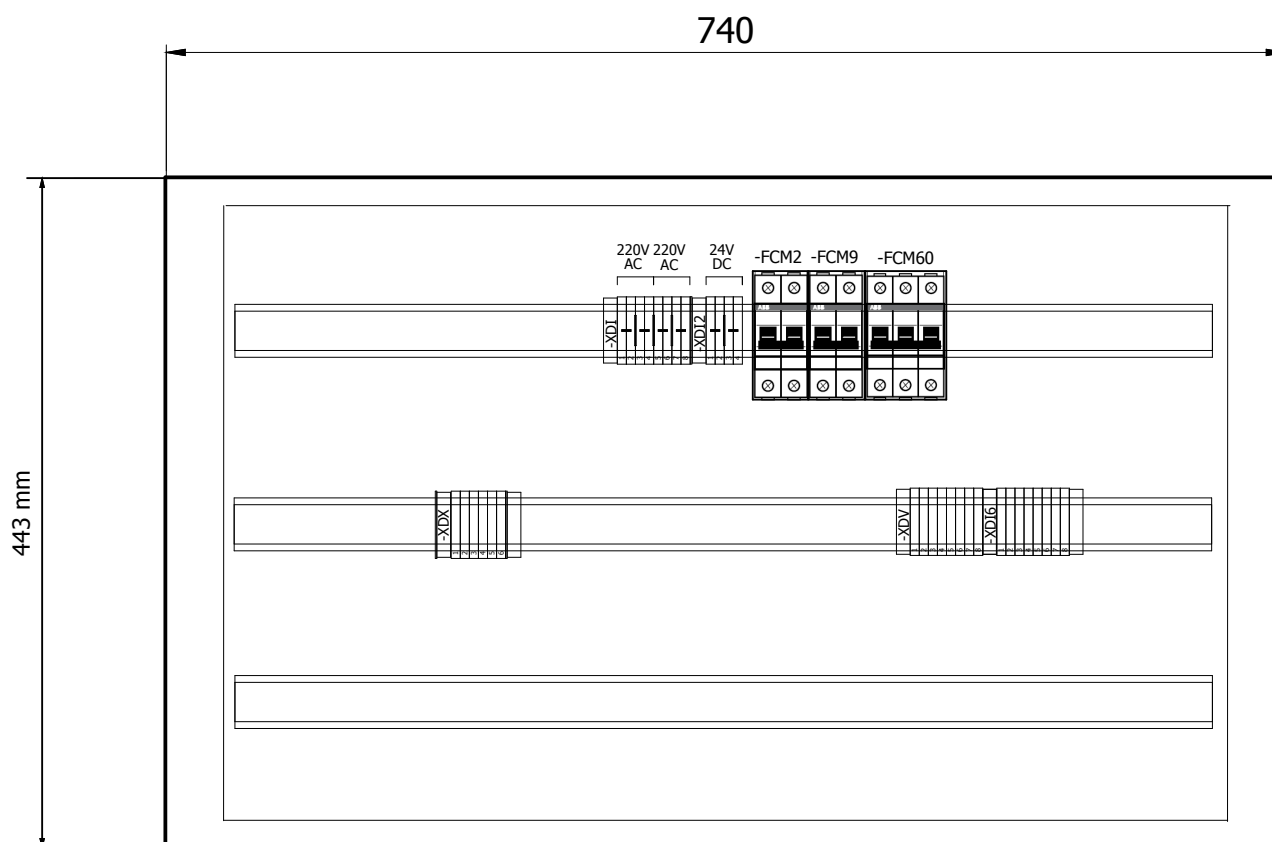
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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant			LV BOX LAYOUT MVD-10&MVD-11 20kV Centralized MVD Room SCHEMATIC DIAGRAM DRS RISER WITH VT	Scale		= H10
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 309
00	Issued for Approval	28.12.18	YR	Checked	AW		Drawing Nbr.	IDUNISEC-BE318042-301		Sh. 10		
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042	Customer Doc. Nbr.			Cont. 16	
						PT. ABB Sakti Industri						

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1		2		3		4		5		6		7		8	
Summarized parts list															
IDABB_F02_005-IDABB															
Carry over: 0.00															
TAG ID		DESCRIPTION				MANUFACTUR/BRAND		PART NUMBER		QUANTITY		REMARKS			
-EB		HEATER 230VAC, 50W				SANYANG		CDXN010033P0009		1		/7.7:E			
-FCM2		MCB S202-C2 MCB. 2P, 2A				ABB		2CDS252001R0024		1		/7.2:C			
-FCM9		MCB S202-C6 MCB. 2P, 6A				ABB		2CDS252001R0064		1		/7.6:C			
-FCM60		MCB S203-C6 MCB. 3P, 6A				ABB		2CDS253001R0064		1		/8.4:C			
-PGV		Digital AC Voltmeter Model : DPM PGD3V, 48 X 96 MM				RISHABH		DPM PGD3V		1		/7.2:D			
-XTV1		Voltage Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting				YONGSUNG		YSPTT-04C		1		/8.5:C			

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